

22/09/2025

INTRODUCTION - W2

DAVID Mihnea-Stefan
NumCS 2025

ABOUT ME - DAVID Mihnea-Stefan

INFORMATIONS

- Fifth semester - BSc CS at ETH Zurich
- TA - AnD, AnW + Coding Assistant D-MATH ETH (i.e. ...)
- University of Pennsylvania (UPenn)
- Soon: FernHagen - Political and Educational Sciences
- Background & Education

HOBBIES & INTERESTS

- Math & Algorithms & Theoretical CS & Physics
- Psychology, Sociology, and Politics
- Educational Sciences in the Pre- and University

Contact Informations

- Email: mihdavid@ethz.ch
- WhatsApp : next Slide
- Instagram: [@david_mihnea](https://www.instagram.com/david_mihnea)

WhatsApp Group : NumCS-G1-2025



Why do we need it?

- Informal communication environment
- Questions and answers (both from me and from you)
- A good way to get to know each other better
- Faster communication
- Homework 🐱

Organization

TA session: Monday, 10:15 – 12:00, CLA E 4

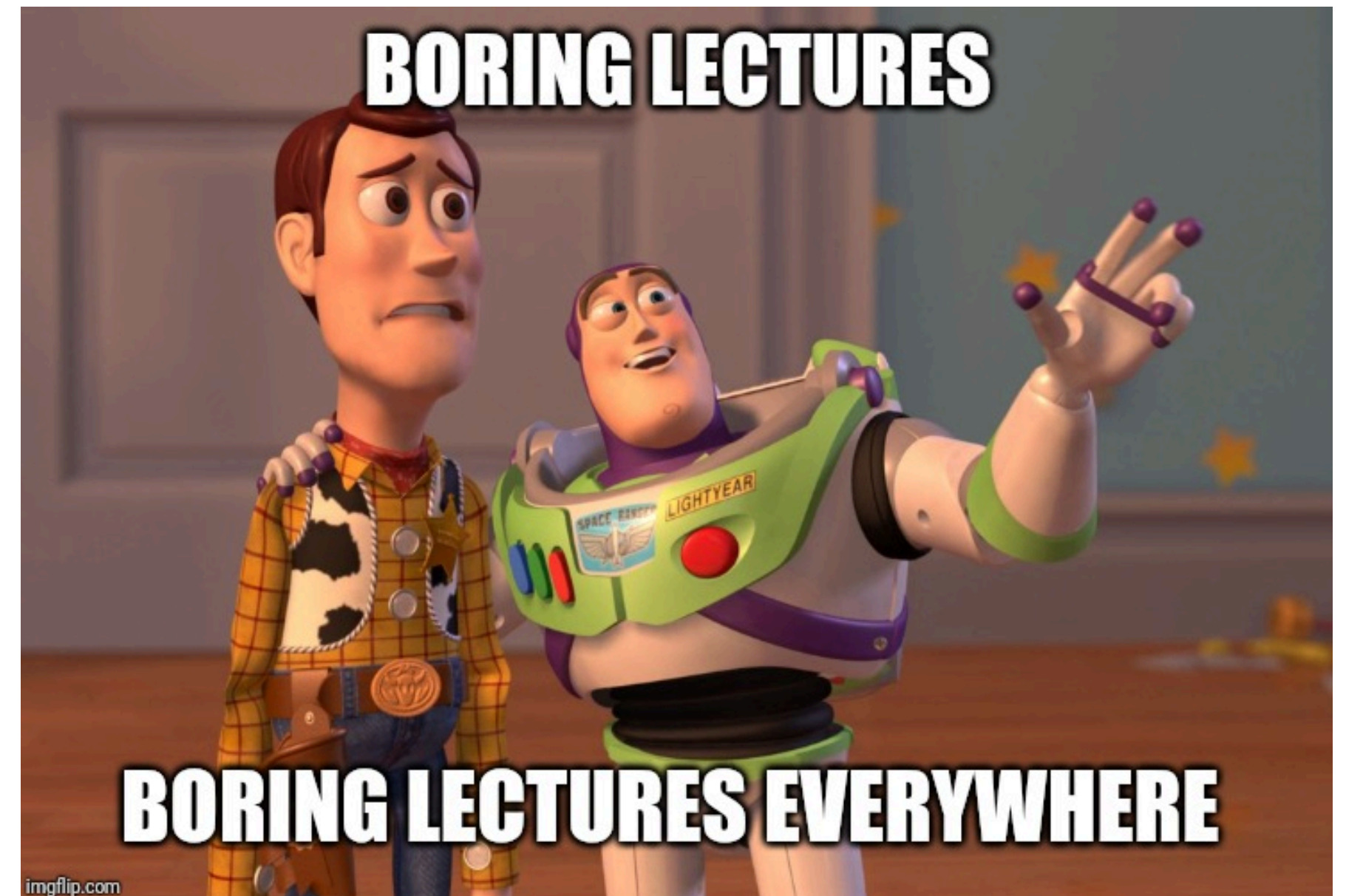
Lectures:

- Monday, 08:15 – 10:00, HG F 1
- Thursday, 10:15 – 12:00, HG F 1

Format: in person (physical attendance)

Grading

- Final exam: 180 minutes
- Computer-based (Moodle)
- Possibility to earn 0.25 bonus points (details on next slide)



Bonus Points

Bonus Points (0.25 max)

1. Code Expert exercises → checked by TAs
2. Weekly Moodle quizzes (1 hour, single attempt) → theory-based, automatically graded

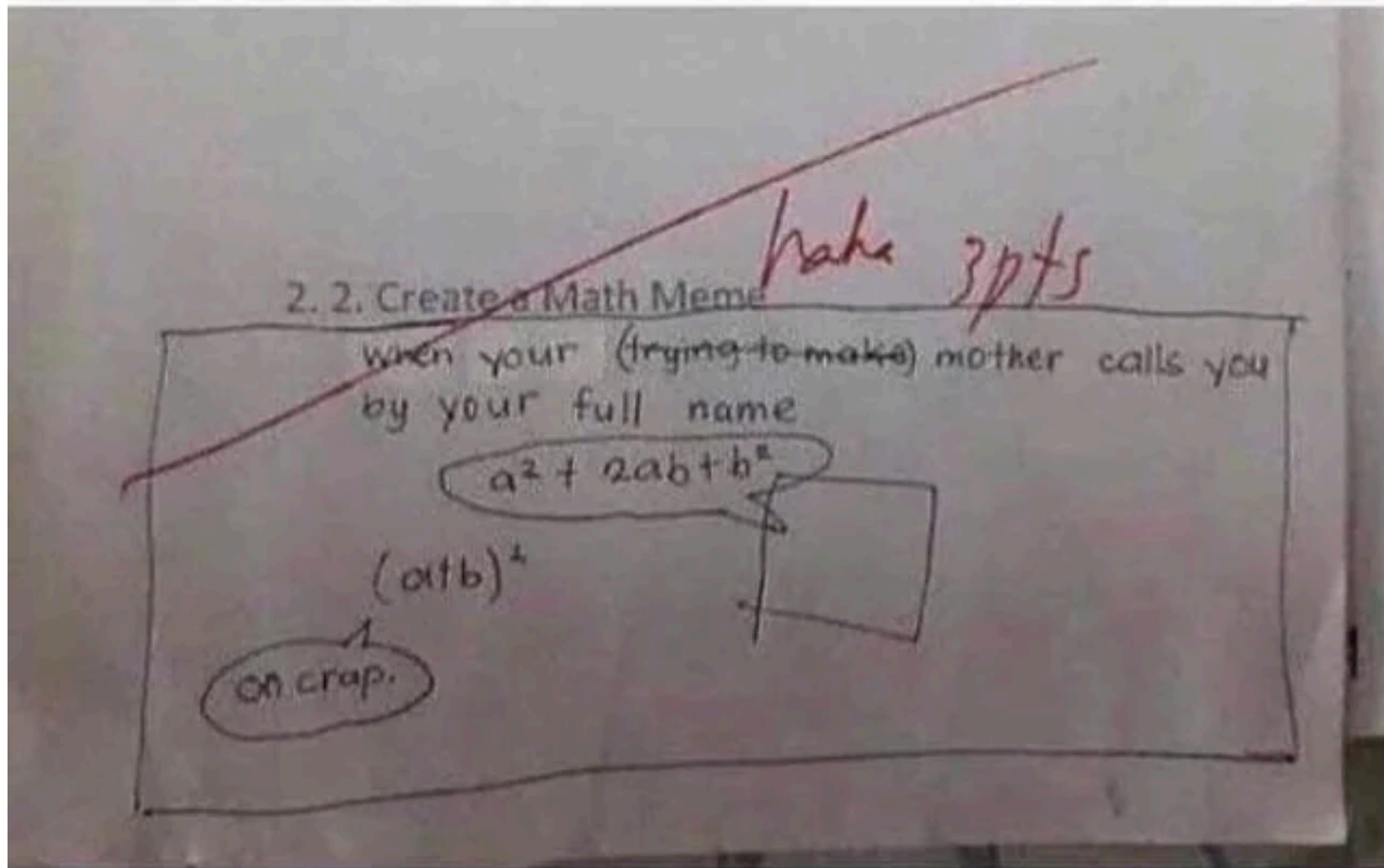
Bonus is not linear:

- $\geq 70\%$ → 0.25 points
- $< 70\%$ → 0 points

HINT!!



Here's a teacher who understands what's important.



Code Expert exercises

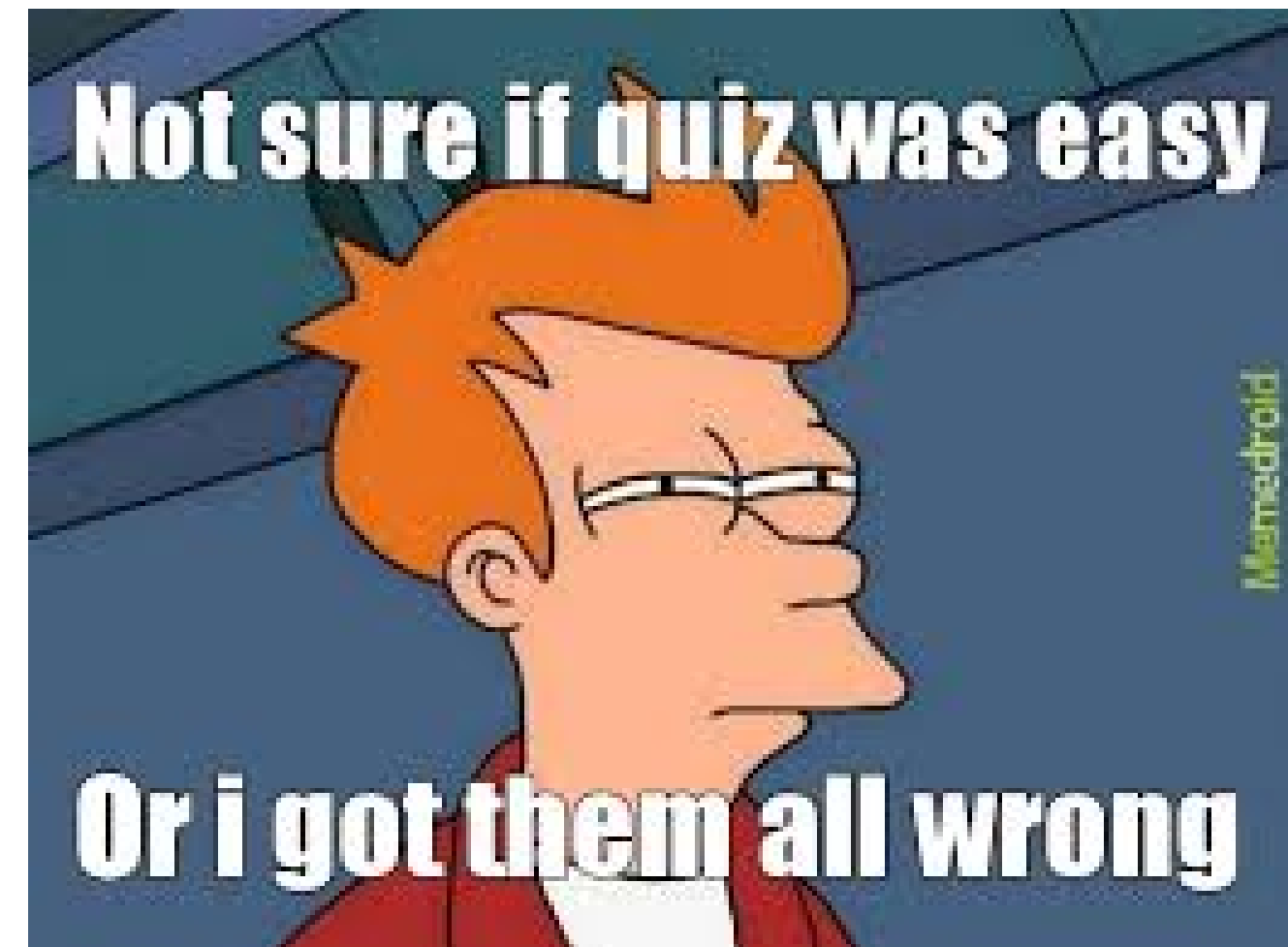
Grading by TAs

I'm usually considered a generous TA 😊
I really appreciate it when students make an effort

So please: even if you don't know the full solution, try something!
That way I'll have a reason to give you partial credit 😊

Weekly Moodle quizzes

- Not difficult 😊
- Be careful: ChatGPT is very bad at solving them 😅
- The answer is always in the script or the notes from that week
- In addition... [*announced only in the exercise session*]



Materials

Every week I will upload:

- **Slides** (if available for that week)
- **Notes** (PDF) → summary of what I consider important for the exam, to make your study process easier
- **Exercises** → highly recommended to solve (I will explain why)



Polybox

ETH zürich

I used to show students something like this in the first exercise session...

Main Topics

1	Continuation of AnD - Graph Properties + P vs. NP in Practice
2	Matching Algorithms + Coloring Algorithms + Their Role in Problem Solving
3	Introduction to Probabilities + Types of Distributions + Classic Problems
4	Randomized Algorithms; Geometric Algorithms
5	Flow Algorithms and Flow Networks

however....

My experience with this course...

What I Can Guarantee....

bonus + exam

Questions?